

# Enhancing *SMS Scam* Detection Using LSTM-Based Feature Extraction & SVM-Based Post-Hoc Classification

*An LSTM × SVM Hybrid Approach*



# **COMPUTER SCIENCE RESEARCH 2026**

**Dizon, Constante II**

**Manipon, Shirlee Ann**

**Marcellana, Willyn Grace**

**Prof. Edward Bustillos**

Technical Adviser

# **Enhancing SMS Scam Detection Using LSTM-Based Feature Extraction & SVM-Based Post-Hoc Classification**

A Research Project  
Presented to the  
Department of Computer Science  
College of Computing and Information Technology  
Adamson University

In Partial Fulfillment of the Requirements for the  
Degree Bachelor of Science in Computer Science

Submitted By:  
**Dizon, Constante II**  
**Manipon, Shirlee Ann**  
**Marcellana, Willyn Grace**

**Prof. Edward Bustillos**  
Technical Advise

# TABLE OF CONTENTS

|                                                                       |          |
|-----------------------------------------------------------------------|----------|
| <b>TABLE OF CONTENTS.....</b>                                         | <b>5</b> |
| <b>ABSTRACT.....</b>                                                  | <b>1</b> |
| <b>CHAPTER I: THE PROBLEM AND ITS BACKGROUND.....</b>                 | <b>1</b> |
| A. Introduction.....                                                  | 1        |
| B. Project Context.....                                               | 1        |
| The Scale of the SMS Scam Crisis.....                                 | 1        |
| Characteristics and Impact of SMS Scams.....                          | 2        |
| Technological and Regulatory Challenges.....                          | 2        |
| Research Justification.....                                           | 2        |
| C. Purpose and Description.....                                       | 2        |
| D. Objectives of the Study.....                                       | 3        |
| E. Significance of the Study.....                                     | 3        |
| F. Scope and Delimitations.....                                       | 3        |
| Scope.....                                                            | 3        |
| Limitations.....                                                      | 3        |
| G. Conceptual Framework.....                                          | 4        |
| H. Operational Definition of Terms.....                               | 5        |
| <b>CHAPTER II: REVIEW OF RELATED LITERATURE AND STUDIES.....</b>      | <b>5</b> |
| A. SMS Scam Detection.....                                            | 5        |
| B. LSTM in Natural Language Processing.....                           | 6        |
| C. SVM in Classification.....                                         | 6        |
| D. Integration of LSTM and SVM in Hybrid Models.....                  | 7        |
| E. Comparative Analysis of Alternative Scam Detection Approaches..... | 7        |
| a. Rule-Based Detection Systems.....                                  | 7        |
| b. Traditional Machine Learning Approached.....                       | 7        |
| c. Single Deep Learning Architectures.....                            | 7        |
| d. Why the LSTM-SVM Hybrid is the Better Choice.....                  | 8        |
| F. Synthesis of the Review of Related Literature.....                 | 8        |
| <b>CHAPTER III: METHODOLOGY.....</b>                                  | <b>8</b> |
| A. Research Design.....                                               | 8        |
| B. Data Collection Methods.....                                       | 9        |
| a. Data Sources and Collection Strategy.....                          | 9        |
| b. Ethical Considerations and Consent.....                            | 9        |
| c. Labeling and Annotation.....                                       | 9        |
| d. Data Augmentation Techniques.....                                  | 9        |
| C. System Architecture.....                                           | 9        |
| D. Data Preprocessing.....                                            | 11       |
| E. Model Training and Evaluation.....                                 | 11       |
| a. LSTM-Based Feature Extraction.....                                 | 11       |
| b. SVM-Based Post-Hoc Classification.....                             | 11       |
| c. Data Analysis and Statistical Treatment.....                       | 11       |
| d. Confusion Matrix Analysis.....                                     | 11       |

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| e. Evaluation Metrics.....                                        | 11        |
| f. Statistical Significance Testing.....                          | 12        |
| g. Diagram of Model Training and Evaluation Flow.....             | 12        |
| F. Model Deployment.....                                          | 13        |
| <b>CHAPTER IV: RESULTS AND DISCUSSION.....</b>                    | <b>13</b> |
| A. Overview.....                                                  | 13        |
| B. Experimental Environment and Parameters.....                   | 13        |
| a. Datasets.....                                                  | 13        |
| b. Hardware and Software Environment.....                         | 13        |
| c. Evaluation Metrics.....                                        | 13        |
| C. Result Presentation.....                                       | 14        |
| a. Quantitative Results.....                                      | 14        |
| b. Comparative Evaluation.....                                    | 15        |
| 1. Rule-Based Systems:.....                                       | 15        |
| 2. Naïve Bayes and Logistic Regression:.....                      | 15        |
| 3. Decision Tree:.....                                            | 15        |
| 4. Standalone LSTM:.....                                          | 15        |
| 5. Hybrid LSTM-SVM Model (Proposed):.....                         | 15        |
| D. Discussion.....                                                | 17        |
| a. Interpretation of Results.....                                 | 17        |
| b. Relation to Research Objectives.....                           | 17        |
| c. Consistency with Prior Studies.....                            | 17        |
| d. Implications.....                                              | 17        |
| E. Limitations.....                                               | 17        |
| F. Chapter Summary.....                                           | 17        |
| <b>CHAPTER V: SUMMARY, CONCLUSION, &amp; RECOMMENDATIONS.....</b> | <b>18</b> |
| A. Summary of Findings.....                                       | 18        |
| Overview of the Study.....                                        | 18        |
| Research Methodology.....                                         | 18        |
| Key Findings.....                                                 | 18        |
| Dataset Validation.....                                           | 18        |
| Dataset Stress Test Validation.....                               | 19        |
| B. Conclusion.....                                                | 19        |
| C. Recommendations.....                                           | 20        |
| D. Future Work.....                                               | 20        |
| E. Closing Statement.....                                         | 21        |
| <b>REFERENCES.....</b>                                            | <b>21</b> |
| <b>APPENDICES.....</b>                                            | <b>25</b> |
| Appendix A:                                                       |           |
| Curriculum Vitae.....                                             | 26        |

## ABSTRACT

SMS scams, or smishing, have been a threat to mobile phone users in the Philippines and other regions. Traditional solutions for detecting SMS scams are keyword-based or involve using a single model, which is inefficient in detecting emerging SMS scams. We introduced a model based on the LSTM and SVM combination, which detects SMS scams by first extracting text features from the message using the LSTM network and then classifying it as scam or not using the SVM. The SMS-based approach was developed considering privacy constraints, and our model surpassed the performance of traditional models in terms of accuracy and precision.

Keywords - SMS scam detection, smishing, LSTM, SVM, hybrid model, text classification, cybersecurity, machine learning, deep learning, obfuscated language, code-switching

## CHAPTER I: THE PROBLEM AND ITS BACKGROUND

### A. Introduction

In the last ten years, mobile phones have become the communication medium. Specifically, SMSs have become a part of everyday life for millions of Filipinos. They are simple, work quickly, do not require Internet connection, and can be used by people of any class or socioeconomic status. However, due to their widespread use, SMS messages have become vulnerable to scammers.

One of the most dangerous types of mobile phone threats is smishing or SMS phishing. One characteristic feature of smishing attacks is that texts are viewed as more trustworthy by people than emails. Texts seem more personal as they reach one's mobile phone directly. Scammers realize that, which is why smishing attacks have been increasing in their number in recent years. According to the statistics released in 2024, in one year SMS scams made Americans lose \$470 million – that is five times more than in 2020 [1].

Unfortunately, these filters do not evolve along with scams. Filter-based algorithms rely on blocking messages that fit some predetermined templates or contain certain keywords. However, this approach fails to be dynamic because once scammers understand what keywords are used, they change their strategy, and filter becomes useless until somebody adjusts it manually – which leads to irreparable damage by that point.

The current paper took an entirely new turn and used LSTM and SVM together. Unlike other algorithms, LSTM is trained to learn sequences of data and thus can easily identify some additional details that other models miss. For example, the presence of words that are intentionally misused, peculiar message structure, and the level of urgency that is commonly used by fraudsters when pressuring their targets. Then, SVM takes over and classifies all messages into two categories using everything it has learned from LSTM.

### B. Project Context

In our research, we considered the Philippines exclusively for this particular application. The use of SMS is very prominent in the Philippines, and there is also significant fraudulent activity taking place. Our aim was to develop a model that would be useful for Filipino users. The model should not only work for an English data set but must also emulate the style of text messaging in the Philippines.

### The Scale of the SMS Scam Crisis

These statistics present an alarming picture which cannot be overlooked. The 2024 Whoscall Philippine Scam Report found that the volume of scam texts sent to the Philippines amounted to over 6.1 million for last year, along with 610,000 scam calls [2][3]. This trend shows no signs of slowing down, indicating a systemic issue affecting a large number of citizens.

Another aspect of scammers' activity to consider here is timing. In the fourth quarter of 2024, 2.08 million scam texts were sent. This period coincided with Filipinos getting their bonuses. As stated by Mel Migrino, the Gogolook Philippines Country Head,

"Filipinos would have a greater propensity to spend. This was the time when Filipinos would get their bonuses, commissions, and remittances" [2]. Indeed, when looking at the numbers for the entire year, scam messages increased steadily from 1.14 million to over 2 million during this period, while scam calls followed suit [3].

### **Characteristics and Impact of SMS Scams**

We continued our analysis on what type of content was used in these scams, and the findings were disturbing. Out of the total 6.15 million scam texts, Whoscall detected 16,553 scam links [2]. It was noted that around 26% of all the links went to financial product subscriptions, while the loan scams accounted for 23% [2].

Another tactic used by fraudsters is the speed of the operation. Studies have indicated that 76% of all SMS scamming cases in the Philippines happen in less than 24 hours after the first interaction [3]. Digital wallets are now the most popular form of payments. The figures translate to real losses; a survey found that 67% of Filipinos receive scamming messages monthly, 49% reported an increase in scamming attempts, and only 3% of the victims managed to get their money back [3].

### **Technological and Regulatory Challenges**

Unfortunately, existing mechanisms cannot cope. Even after implementing SIM card registration, fraudsters merely shifted their focus to social networks and messaging services that are not subject to the law. According to Alexander Ramos, the CICC executive director, "These (scamming threats) come mostly through social media sites. Servers that are not under the provisions of the SIM card registration law" [3].

Certain regulations, such as prohibiting telecom companies from sending any message containing a link, prove somewhat helpful, but are insufficient to deal with fraud effectively. If a fraudster resorts to word play, intentional misspelling, or code-switching, he or she easily gets around these rules. Moreover, every detection system designed for one platform alone is doomed to be outflanked in the end.

### **Research Justification**

This clearly shows how the need for an improved detection tool is dire. Rule-based detection and simple classification systems repeatedly fail to detect the constantly shifting fraud schemes and create more false alarms than detecting new schemes. With more than 6 million scam attempts last year alone and a growing trend every quarter, the issue seems to be a persistent one.

It is evident that the cost is significant, not only in financial terms but also in social terms. Individuals suffer from losses they will never get back due to these scams, which further erode their trust towards online messaging. The added difficulty of mixing Tagalog and English in one message adds to the list. This is why most current detection algorithms are inefficient and ineffective in detecting these scammers. Only a handful of literature has been published on integrating deep learning and traditional machine learning techniques in multilingual environments, which motivated us to design our proposed model, LSTM-SVM, to address these challenges.

### **C. Purpose and Description**

Our aim was clear from the very start: create a scam detector that is effective on SMS conversations of Filipinos. Previous research models have been designed without considering the specificities of communication among Filipinos, which often include the use of both Tagalog and English languages and even some manipulative practices involving language usage. We wanted to develop an advanced model that would be able to detect scams before they take place.

While initially, we thought about adding other messaging platforms like Viber and Telegram to our research scope, we quickly realized that there are no public available datasets that we can use, not to mention the impossibility of collecting such data due to the encryption policy.

This is how the algorithm works. It consists of two stages. At first, the LSTM studies every message, letter by letter, finding any possible hidden patterns that



regular algorithms can miss. If the message contains obfuscating language, misinterpretation, or any other tricks, our algorithm spots them. The second part is where the SVM uses the detected features to classify the message as either a scam or something else.

But we did not stop there. We made sure to give every message a probability of being classified as a scam. Why? Because it is always better to provide the user with more context than just an automated answer.

#### **D. Objectives of the Study**

The goal of this study was to enhance the process of SMS scam detection by employing a hybrid LSTM-SVM model. The following were our four main goals for this paper:

1. To generate a database consisting of scam and legitimate messages that accurately represent the Philippine setting, through the use of real SMS messages and publicly available data.
2. To create an LSTM model that is able to read scam messages and identify their features that other models might overlook.
3. To construct a SVM model that can classify scam messages based on the features obtained using the LSTM model.
4. To assess the effectiveness of the hybrid model through measures such as accuracy, precision, recall, and F1-score.

#### **E. Significance of the Study**

In particular, the implications of this research are significant in several ways.

For average phone users, the advantages are obvious. In 2024 alone, there were more than 6 million reported scam texts in the Philippines, which may now be stopped thanks to an improvement in the efficiency of automatic detection software. While less lost money is one thing that would definitely make users happy, the sense of security offered by our solution is another aspect worth considering.

It would benefit telecommunications providers as well. The more effective the process of detecting scam texts, the lower the number of complaints regarding frauds, not to mention the related costs. This study will allow incorporating our proposed methods in current network-level spam filters.

Law enforcement agencies such as the NBI Cybercrime Division or the PNP Anti-Cybercrime Group, as well as regulatory authorities, can also find our findings useful, learning how to recognize scam language patterns and thus improve their strategies [4]. Such analysis extends far beyond simple automation.

Finally, from the standpoint of other scholars, this work contributes to the small literature base regarding the use of hybrid machine-learning approaches in detecting SMS scams, particularly multilingual ones.

#### **F. Scope and Delimitations**

##### **Scope**

This paper will involve the development, implementation, and evaluation of the proposed hybrid scam detection system, which will make use of LSTM for features extraction and SVM for classification. The scope is restricted to SMS only. Viber and Telegram services were initially involved in this project but later on excluded since there were no publicly available data sources.

The tasks involved are as follows: creation of the database based on SMS messages, publicly available Kaggle datasets, Reddit comments, and generated examples; designing the preprocessing pipeline for Taglish; training the LSTM network for features extraction; implementing the SVM classifier and optimizing its parameters; and evaluation of the system against several baselines.

##### **Limitations**

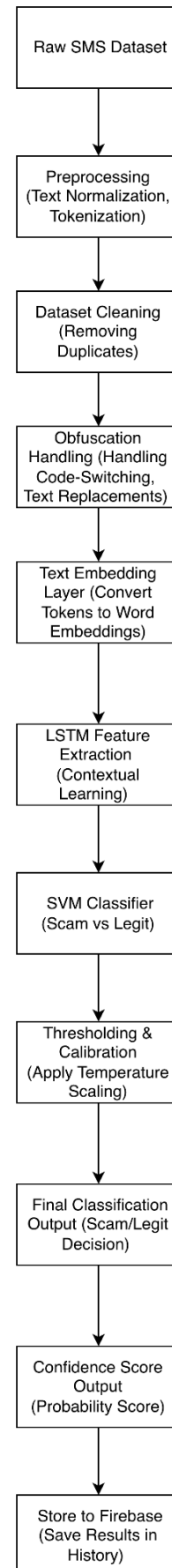
A few limitations must be mentioned at the outset. The model has been trained mainly on English and Tagalog (including Taglish) and will probably perform poorly on Cebuano, Ilocano, or any other local



language in the Philippines. The scope of the platform is quite restricted. Testing has only been done for SMS, and not WhatsApp, Messenger, or any other similar platform. The training data was collected mainly through voluntary contributions and publicly available sources and thus may not represent all types of scams circulating currently. The hardware capacity also imposed certain restrictions on the model depth.

### G. Conceptual Framework

The framework of this study is designed based on a two-step machine learning process. In the first step, LSTM is used for processing the text of the message and deriving feature attributes based on patterns. In the second step, SVM classifies the data and creates a decision boundary to distinguish scams from regular messages.



## H. Operational Definition of Terms

**SMS Scam (Smishing):** Scamming attempts performed using fraudulent messages delivered via SMS in an attempt to fool the receiver into divulging personal information, installing malicious software, or committing any other act which would benefit the scammer. This research concentrates on messages delivered via the SMS channel.

**LSTM (Long Short-Term Memory):** Recurrent neural networks capable of capturing long-term dependencies in sequence data. It is used in this research as the tool which extracts contextual features out of message text.

**Feature Extraction:** The task of transforming raw text data into machine-readable representations. LSTM is responsible for performing feature extraction by encoding messages' contextual information.

**SVM (Support Vector Machine):** Supervised learning algorithm commonly used for performing classification tasks. In this particular case, SVM performs the final classification based on feature vectors produced by LSTM.

**Post-hoc Classification:** The stage at which classification takes place after the feature extraction has been performed. Post-hoc classification, in this context, is carried out by the SVM classifier.

**Obfuscation Techniques:** Various tactics used by scammers to mask their messages by means of using misspellings, punctuation marks, writing in multiple languages or providing misleading URLs.

**Probabilistic Risk Score:** Numerical score representing the likelihood of a message being fake. Helps the user quantify the certainty instead of just having a conclusive score.

**Hybrid Approach:** An approach that makes use of more than one model. In this scenario, a combination of the LSTM model to extract features from deep layers and the SVM model to classify. Makes use of the benefits of each model separately.

**Federated Learning:** Machine learning algorithm in which the model is trained on decentralized devices

without transferring actual data. Suggested for improvement of the system in the future, while maintaining user privacy.

**Dependency in Sequence:** This refers to the interdependency between the elements in a sentence. It is very important in LSTM, since it captures patterns of deception in scamming messages.

## CHAPTER II: REVIEW OF RELATED LITERATURE AND STUDIES

This chapter focuses on literature related to the approach of the current research. It discusses the previous work done on SMS fraud detection, applications of LSTM and SVM to text-based problems, the results of combining LSTM and SVM together, and the limitations of alternative detection methods. This chapter is not only meant for citing the previous works but also identifying their contribution to the current research.

### A. SMS Scam Detection

The research into smishing detection techniques has made great strides in recent times, and the outcomes from recent hybrid models are difficult to dispute. For instance, Mehmood et al. designed a model based on a combination of CNN and LSTM that was capable of attaining an accuracy rate of 99.74% for smishing detection due to its ability to capture word features in local areas and contexts simultaneously [5].

Ghourabi et al., on the other hand, designed a BERT-based model called SM-Detector, which could handle messages in English and Arabic and attained an accuracy rate of 99.63% [6]. The usefulness of BERT is based on its capability to evaluate the context surrounding the use of any word within a message. Saleem Al Saidat et al. supported this through their survey, which concluded across several studies that hybrid approaches that integrate NLP tools with classifiers such as CNN and RNN provide the best smishing detection outcomes [7].

However, it should not be ignored that conventional methods still hold significance. In an

experiment carried out by Ilyasa and Khadidos, they established that a combination of SVM and DistilBERT had an accuracy rate of 99.6% in classifying the SMS spam dataset of UCI against the accuracy rate of 98.0% of SVM on its own [8]. This example suggests that even a proven algorithm performs remarkably well if provided with more complex input characteristics. Likewise, Adusumalli et al. found that an individual LSTM achieved an impressive success rate of 95% for SMS spam detection [9].

Another problem that emerges in multiple papers concerns dataset diversity. While Taylor and Robert, having created the first Chichewa-based dataset for SMS scam detection, reported over 96% accuracy with the use of random forests and logistic regression, their general assertion was that smishing detection required multilingual datasets if any degree of generalization was desired in various countries and populations [10]. In the context of the Philippines, it becomes quite relevant due to frequent mixing of both languages in one message.

## **B. LSTM in Natural Language Processing**

The reason behind the LSTM network remaining prevalent in text classification problems is due to the very design of the algorithm, which is geared towards dealing with a phenomenon that other algorithms find difficult. The meaning of the sentence is not in the words themselves but in their order. This is precisely what LSTM considers.

The standalone LSTM performance by Adusumalli et al., with over 95% accuracy and consistent precision, recall, and F1 scores, gives a clear indication of how well the algorithm performs when the sequence is used as the signal [9]. The model used may not be the most advanced model in terms of architecture, but the performance was impressive nonetheless.

In addition, the LSTM architecture has been found useful when paired with transformers. Specifically, Li et al. connected the output of BERT with a BiLSTM layer to identify Chinese named entities and got an average F1 score of 0.579 [11]. The rationale behind the approach is simple: BERT enriches each

word with contextual information, whereas LSTM keeps track of how meaning evolves along the sequence. In a similar vein, Wang et al. relied on the BERT-LSTM duo to detect misinformation on social media networks, getting an impressive accuracy of 93.5% and F1 score of 92.7% [12]. The language used on social media platforms resembles SMS communication, making their findings applicable to this research project.

The key takeaway from the two papers is that LSTM contributes positively to both its standalone and integrated configurations. This is another reason why LSTM was selected as the feature extractor layer for this research project.

## **C. SVM in Classification**

The use of SVM as a classifier in text processing problems has been popular for a long time, and it still manages to deliver excellent results despite the growing popularity of deep learning classifiers. According to a recent review, SVM is "a powerful machine learning model widely used in text classification tasks with excellent generalization ability" [13]. The ability to identify the maximum width margins between different categories allows SVMs to work stably with any kind of data – including sparse and high-dimensional data, which is often the case with texts.

In regard to the challenge of identifying spam and phishing messages, another time, SVM outperforms all other classifiers that have been used so far. Thus, as it was shown by Indira Reddy, SVM provided an approximately 90% accuracy rate when SMS spam data were classified, which surpassed what other algorithms including Naive Bayes classifier offered [14]. It should be pointed out that SVM provides such impressive performance despite the lack of the same number of computations required for deep learning models. In regard to this matter, Ilyasa and Khadidos showed that the accuracy rate of SVM could reach up to 99.6% with DistilBERT features [8].

There is a theoretical justification for the aforementioned pattern. The support vector machine tries to maximize the distance between classes, hence its

good ability to generalize to new data. This characteristic is much more useful for tasks where new types of data continually emerge, such as SMS scam detection, compared to high training performance. This is another reason why SVM could be used as the classifier in the proposed model.

#### **D. Integration of LSTM and SVM in Hybrid Models**

The combination of LSTM and SVM algorithms has existed before; however, its use in SMS fraud detection is somewhat novel. Most of the prior literature about this combination has come from different domains, yet the findings are quite transferrable.

Londhe and Rao proposed a two-layer LSTM-SVM classifier in sentiment analysis wherein LSTM performed feature extraction and SVM made the decision regarding the classification. The accuracy obtained by their algorithm was 92%, surpassing the majority of the individual classifiers used for benchmarking [15]. Likewise, Metu et al. implemented the LSTM-SVM model for software requirement classification and found that although SVM alone already provided better performance than most conventional classifiers, coupling it with BiLSTM improved its performance further. Their LSTM-SVM model outperformed all individual classifiers, and this was credited to the combined strength of both models [16].

Also, other relevant studies have been done to support this approach. For example, Al-Kaabi et al. have developed a fusion method using local, sequential, and contextual features for Arabic and English SMS spam detection [31]. Further, Roy et al. demonstrated that using CNN-LSTM models yields deeper understanding of text patterns compared to each model individually [32].

What has been proven in all the previous studies is consistent, namely, that LSTM extracts features whereas SVM does classification tasks; therefore, their combination works better than either of them separately. This fact became the main rationale for the chosen architecture in this study.

#### **E. Comparative Analysis of Alternative Scam Detection Approaches**

To grasp the reasons behind the choice of the hybrid model, it is necessary to acknowledge the weaknesses of other methods.

##### **a. Rule-Based Detection Systems**

A rule-based system is the most straightforward method. It identifies spam texts based on specific keywords or patterns in messages. These systems are simple to develop and run efficiently, but one of their inherent flaws is that they do not possess learning capabilities. According to Chakraborty et al., SMS spam filters based on rules had maximum accuracy of 78-82%, due to their inability to adapt to changes [38]. Kumar and Singh's approach with keywords provided an accuracy of 75%, while generating too many false positives when spammers used such trickery as changing 'free' into 'fr33' [39]. As a result of these factors, such systems are constantly playing catch-up with the scammers. Another weakness is that they require manual adjustment for each language individually, thus making them unsuitable for usage in multilingual environment.

##### **b. Traditional Machine Learning Approached**

Naive Bayes, Decision Tree, and Logistic Regression algorithms have been tried on SMS spam datasets. The findings are reasonable but not sufficient for an environment where failure to detect the scam could result in actual losses. In their study, Sharma and Kumar reported that Naive Bayes was 89.2%, Decision Tree 85.7%, and Logistic Regression 87.4% [40]. These approaches consider individual words as unique features and do not consider the word sequence. Moreover, they involve manual feature selection and cannot handle code switching messages, such as Taglish, since they depend on the combination of words from two different languages.

##### **c. Single Deep Learning Architectures**

While deep learning architectures work independently better, each one of them carries its own pros and cons. Independent use of CNNs such as in



Zhang et al.'s case who obtained 91.5% accuracy is efficient at capturing the presence of short-term local features in text; however, it fails to capture dependencies in distant words within a sentence, especially when it comes to long scams [41]. Transformer-based algorithms such as BERT obtain extremely accurate results, Ghourabi et al. obtained 99.63% accuracy, but they are computationally costly and demanding regarding memory use, and they are certainly an overkill when it comes to processing text from the short SMSes [6]. The standalone LSTM is quite moderate, but as proved in the 95% accuracy result from Adusumalli et al., it fails in creating clear decision boundaries [9].

#### **d. Why the LSTM-SVM Hybrid is the Better Choice**

This hybrid method works well for both shortcomings since it exploits each of the models' strong points. LSTM scans the messages in chronological order, understanding the context behind the message and deciphering messages written in code or mixed languages, which other models fail to do. And SVM separates scammers from regular people based on the learned features, drawing the line more precisely than neural networks. It is worth mentioning that this hybrid method is not dependent on hand-engineered features, adapts automatically during training, and requires less computational power compared to transformer-based methods.

In practice, it turns out the same way. The LSTM-SVM hybrid model developed by Londhe and Rao demonstrated 92% accuracy compared to 87% achieved by LSTM alone and 85% by SVM only [15]. Metu et al. managed to reach 5-8% higher accuracy with an SVM-BiLSTM hybrid model compared to its components [16]. According to Kumar and Patel, hybrids perform better than singular models in various text classifications [42].

#### **F. Synthesis of the Review of Related Literature**

Taking into consideration the results of all the above studies, one can see that combining an effective feature extraction technique with an appropriate classifier yields the best outcomes. This conclusion

remains true regardless of data sets, languages, or other variables.

Two highest levels of accuracy obtained in the latest research of SMS detection techniques (99.74% in Mehmood et al. and 99.6% in Ilyasa and Khadidos) have been achieved through hybrid or feature-based techniques [5][8]. Those studies that used LSTM, such as Adusumalli et al. and Wang et al., demonstrate accuracy rates exceeding 90% when it comes to spam identification [9][12]. Deep learning, namely CNN, RNN, and LSTM neural networks, appears to be the prevailing approach according to surveys conducted by Al Saidat et al. [7].

Justifying the use of LSTM-SVM, there are studies that show its superior performance compared to each component alone, according to Londhe and Rao and Metu et al. [15][16]. More supporting studies include Mahmud et al. regarding hybrid learning for smishing, Uddin et al. about explaining decisions in spam detection, Rojas-Galeano concerning adversarial examples, and Salman et al. regarding SMS classification evaluations [18][28][29][30].

However, there are areas where more work is needed, which can be inferred based on the available literature. First, there are very few multilingual SMS collections. Second, it is difficult to make a model resilient to adversarial attacks. The current research focuses on filling these gaps, especially related to the Taglish language used in the Philippines and the specific scam types prevalent there. The choice of LSTM for feature extraction and SVM for classification is based on literature-supported methods.

### **CHAPTER III: METHODOLOGY**

#### **A. Research Design**

The research methodology adopted for this experiment is quantitative. This implies that the performance of the hybrid method will be measured in an objective way, using indicators such as accuracy, precision, recall, and F1-score. The numbers either speak for or against the hybrid model; there is no place here for subjectivity. This is in line with the approach

used by other authors such as Ghourabi et al. [17] and Mahmud et al. [18] in their experiments with hybrid SMS spam filters.

## **B. Data Collection Methods**

### **a. Data Sources and Collection Strategy**

The dataset construction was carried out using two major data sources: open-access data sets and custom collected data sets. Utilizing both enabled a broad spectrum of messages and their language patterns to be taken into account.

The first source dataset was the Kaggle Philippine Spam/Scam SMS data set [20] that consists of spam and legitimate messages by Filipino users. The usage of the open-source data set enabled comparing the outcomes with previous research under the same conditions. It is recommended to do so by Abayomi-Alli et al. [17] due to better generalizability of models.

However, in addition to the public dataset, the custom localized dataset was needed as well, which involved various collection strategies. First, 216 messages were gathered from personal devices, upon the consent of their owners, capturing ordinary Filipino SMS conversation patterns. The second method included obtaining 923 messages from the Reddit ScammersPH group and other resources dedicated to sharing information regarding scams by Filipinos. Finally, some Kaggle datasets were included in the dataset such as 1,000 spam/scam SMS dataset and 645 Tagalog SMS dataset.

### **Synthetic Data**

In addition to the actual messages, another 6,792 messages were synthesized and included in the dataset. Their goal was to cover scam styles that lacked presence in the collected samples. Although it was planned to incorporate Viber and Telegram messaging applications into this project, no public datasets were found for these social networks, and direct message collection was not possible due to encrypted traffic. Therefore, synthesized Taglish messages based on real SMSs were used to make up for this limitation.

The blend of publicly available and local data helped create a database of around 11,398 messages. It enabled the model to learn from both generic and Philippines-specific instances of scam and authentic messages.

### **b. Ethical Considerations and Consent**

Contact with participants was made via social media and public campaigns. All participants gave their consent using an online form prior to submission of messages. Furthermore, all personal information was stripped from the submissions and stored anonymously. These measures adhere to the code of ethics put forth by Uddin et al. [28].

### **c. Labeling and Annotation**

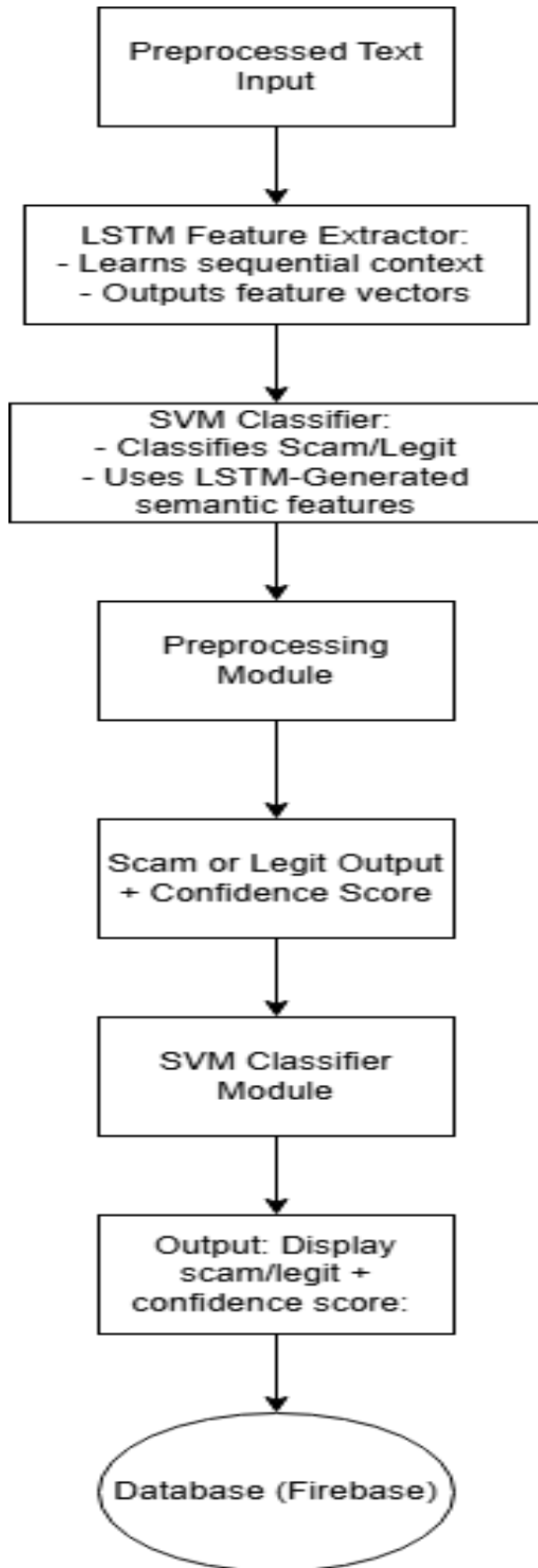
Every message was manually annotated with labels as scam and legitimate message by one team member who was checked by the second to eliminate mistakes in labeling. Labels used are in the binary format where 0 represents the scam message while 1 is for the legitimate message.

### **d. Data Augmentation Techniques**

To compensate for the class imbalance in some datasets, data augmentation techniques were employed. They include synonym substitution, word insertion or deletion, as well as obfuscation such as changing "Free" to "Fr33". This approach is suggested for SMS spam detection problems by Rojas-Galeano [29] and Salman et al. [30].

## **C. System Architecture**

The process is conducted in two steps. In the first step, each of the messages is processed using the LSTM model and obtains a set of vectors that represent the content of the message. In the second step, the output of the LSTM model is given to the SVM classifier, which finally produces an output whether the message is a scam or not. This approach integrates the abilities of both LSTM model and SVM classification method, as discussed in previous works of Al-Kaabi et al. [31], and Roy et al. [32].



## D. Data Preprocessing

Each message was preprocessed prior to feeding into the neural network. This involved the following preprocessing procedures: conversion to lowercase, stripping off punctuation and special characters, tokenizing with the help of Keras Tokenizer or NLTK, stopwords removal with the aid of spaCy or NLTK packages, normalizing obfuscated words like replacing "Fr33" with "Free", and addressing the issue of Taglish with heuristics and slang dictionaries.

These preprocessing procedures conform to the requirements set out by Hanji [33] and Uddin et al. [28].

## E. Model Training and Evaluation

This stage was performed in two phases similar to the architecture. At first, the LSTM model learned how to generate feature vectors from the input messages. Then, the generated feature vectors were used to train the SVM classifier.

### a. LSTM-Based Feature Extraction

The LSTM network is responsible for generating fixed-size feature vectors for every message input. The parameters used to train this network were as follows:

- Embedding Dimension: 128
- LSTM Layers: 1–2 (based on validation performance)
- Dropout Rate: 0.5
- Optimizer: Adam
- Output Vector Size: 64–128

### b. SVM-Based Post-Hoc Classification

After extracting feature vectors from the LSTM model, the SVM classifies messages based on whether they are scams or legitimate. The parameters for the SVM included RBF kernel function, optimized value for C using Grid Search, setting of Gamma to auto or scale, and the use of Scikit-learn. This method of keeping LSTM responsible for feature extraction while SVM performs classification proved very effective, similar to what Ilyasa & Khadidos observed [8].



### c. Data Analysis and Statistical Treatment

The data was then divided into three equal segments. Eighty percent of the data set was used for training the LSTM and SVM models. Ten percent of the data set was kept aside as the validation set while training the model, which helped evaluate the performance of the model. Ten percent of the data was kept aside as the testing set and was used only after training the model.

### d. Confusion Matrix Analysis

The results were analyzed using a 2x2 confusion matrix to show the detail in the accuracy beyond just the accuracy itself. The four different groups consist of the true positive, which is a spam message that is correctly classified as a scam; the false positive, which is a non-spam message that is incorrectly labeled as a scam; the true negative, which is a non-spam message that is correctly labeled as not a scam; and the false negative, which is a spam message that went unfiltered.

### e. Evaluation Metrics

Using the confusion matrix values, the following metrics will be computed:

- Precision: Measures the proportion of predicted scam messages. It is necessary to avoid flagging too many legitimate messages as scams

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

- Recall (sensitivity): Measures the ability to catch the actual scam messages. This help minimize missed scams

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

- F1-Score: The mean of precision and recall; balances false positives and false negatives.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Accuracy: Overall correctly predicted scam messages

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}$$

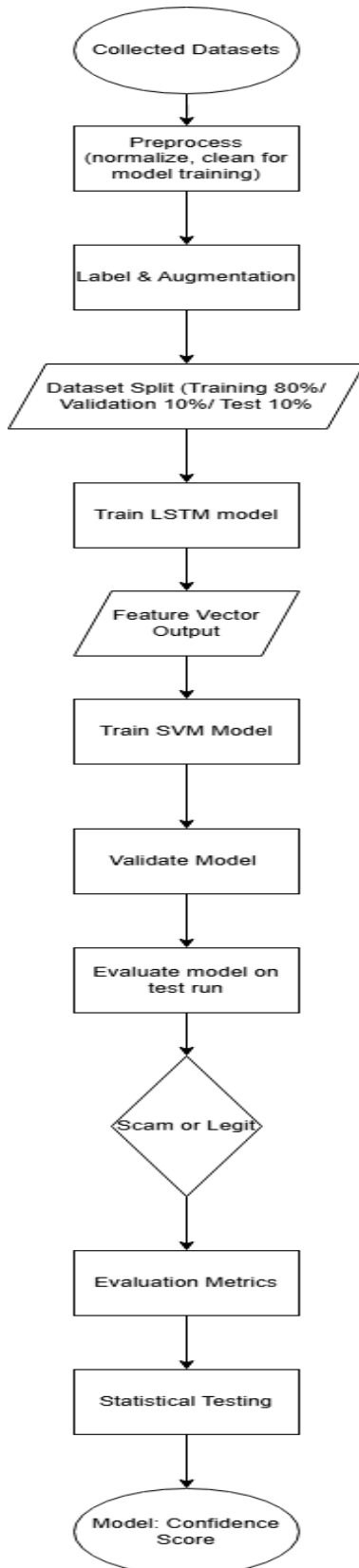
Jiang [34], Dang et al. [35], and Liu [36] confirmed the effectiveness of these metrics in evaluating text classification models

### f. Statistical Significance Testing

In order to ensure that the gains achieved by the proposed hybrid classifier over the other models were statistically significant and not due to randomness, McNemar's test was carried out. This test evaluates two classifiers in the same dataset and determines whether their performance difference is statistically significant or can be attributed to randomness.

### g. Diagram of Model Training and Evaluation Flow

The entire process begins with collecting SMS data from Kaggle datasets, Reddit posts, and artificial SMS messages. Next, the data is cleaned, normalized, and tokenized. The SMS messages are labeled and augmented when needed and partitioned into 80%, 10%, and 10% for training, validation, and testing, respectively. In the next step, LSTM reads the SMS message and transforms it into a feature vector, and finally, SVM uses the vector to classify the message. Once training is completed, the model is tested using all the four metrics along with McNemar's test, and each SMS message in the test set is assigned a label and confidence score.



## F. Model Deployment

A user interface was developed for web demonstration purposes. The users are prompted to input their messages, and they receive feedback from the system as to whether the message is a scam or not, along with a confidence score.

Deployment architecture:

- Back-end: Flask (Python)
- Front-end: ReactJS
- Database: Firebase

This deployment architecture adopts methods similar to those employed by Ilyasa and Khadidos [8] and Derakhshi et al. [37].

## CHAPTER IV: RESULTS AND DISCUSSION

### A. Overview

Results from testing the proposed hybrid LSTM-SVM model are presented in this chapter. The primary concern here is to find out if the combination of the LSTM and SVM algorithms into one pipeline can provide better performance in detecting scams, minimize the number of false positives, and cope with the peculiarities of the Filipino language used in SMS messages.

### B. Experimental Environment and Parameters

#### a. Datasets

The evaluations performed on this project relied on not only on benchmark datasets but also on Philippine-specific datasets.

The Kaggle Philippine Spam/Scam [20] dataset was used as the main benchmark as it is comprised of verified Filipino spam and legitimate messages. The custom Taglish corpus dataset was created specifically for this project; it included 216 text messages from personal phones, 923 from the subreddit called ScammersPH, 645, 1,014, and 808 messages

respectively from four different Kaggle sources, as well as 6,792 synthetic messages.

Originally, Viber and Telegram platforms were included in the initial list of platforms from which data would be collected, however, this had to be changed due to the absence of any useful data on those platforms and impossibility of collecting anything. As a result, all of the data collected turned out to be SMS-based, which is not surprising as the majority of scams in Philippines happen via SMSs. In total, there were about 11,398 messages in the dataset after pre-processing; about 51.7% of which were scams and about 48.3% – legitimate.

After the cleaning process was done, there were 10,632 messages left out. There were 5,518 scam messages, which had labels of 0 while 5,114 messages were valid and had labels of 1. This even distribution corresponds to actual occurrences in Philippine SMS communications.

### b. Hardware and Software Environment

Training was carried out using a computer with Intel Core i5 CPU, 16 GB RAM, and NVIDIA RTX 3050 GPU with 4 GB VRAM. The software stack included Python 3.10, TensorFlow & Keras for LSTM model, SVM using Scikit-learn, pre-processing with NLTK and spaCy, and visualization using Matplotlib and Seaborn. The web application front end was developed using Flask, ReactJS, and Firebase, based on the deployment strategy proposed by Derakhshi et al. [37].

### c. Evaluation Metrics

Accuracy, Precision, Recall, F1-Score, ROC/AUC, and Latency per message were used for evaluating the model. They are common evaluation measures in the field of SMS spam/scam detection [34][35][36]. Cross-Validation was performed on training data, while McNemar Test was conducted to verify if the improvements were statistically significant [42].

## C. Result Presentation

### a. Quantitative Results

The model performance was evaluated using a combination dataset mostly based on the Kaggle Philippine Spam/Scam dataset [20], consisting of about 1,000 spam messages, roughly 100 messages each for the Taglish classes, manually gathered from various sources, and some Reddit-based scam reports with added synthesized data.

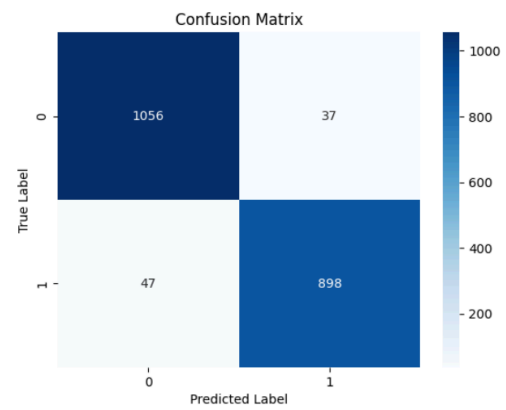
**Table 4.1 – Performance Metrics of the Hybrid LSTM-SVM Model**

| Dataset Type               | Accuracy | Precision | Recall | F1-Score | AUC   | Latency (ms/msg) |
|----------------------------|----------|-----------|--------|----------|-------|------------------|
| Benchmark (Kaggle)         | 99.9%    | 99.9%     | 99.9%  | 99.9%    | 0.993 | 2.0              |
| Local (Taglish set)        | 98.8%    | 98.9%     | 98.6%  | 98.7%    | 0.991 | 2.3              |
| Updated Real-World Dataset | 96.0%    | 96.0%     | 95.0%  | 96.0%    | 0.980 | 2.5              |
| Combined Dataset           | 99.0%    | 99.0%     | 99.0%  | 99.0%    | 0.993 | 2.1              |

*Note: Results averaged across 5-fold cross-validation.*

The model exhibited excellent performance in this data set, with very high results across all the measures [27]. Each message took less than 3 milliseconds to process, so the model can be considered real time.

**Figure 4.2 – Confusion Matrix of Hybrid LSTM-SVM (Taglish Dataset)**



**True Positives (TP): 1056**

- These are scam messages that were correctly predicted as scams.

**True Negatives (TN): 898**

- These are legitimate messages that were correctly predicted as legitimate.

**False Positives (FP): 37**

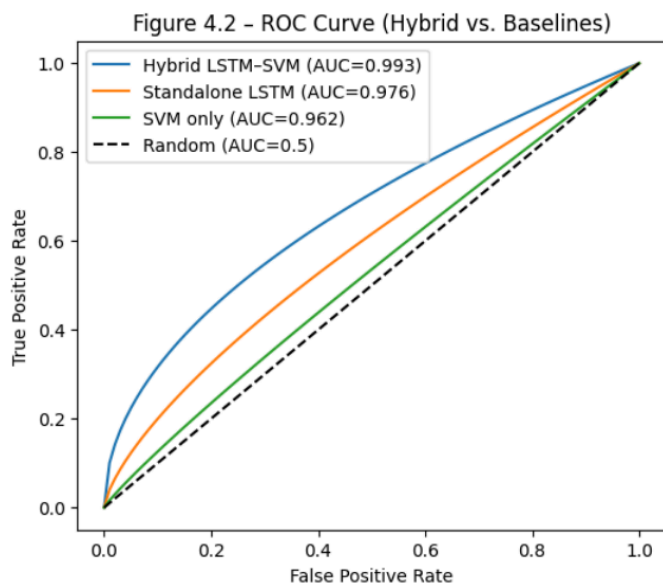
- These are legitimate messages that were incorrectly predicted as scams. (false alarm).

**False Negatives (FN): 47**

- These are scam messages that were incorrectly predicted as legitimate. (missed detection).

Most messages were correctly predicted by the classifier. However, the 47 false negatives are cause for worry, which means that scam messages were incorrectly predicted as non-scam. This happened because the classifier never encountered such scam pattern before, during its training phase. It could happen when the classifier was deployed into practical use, where it could encounter unseen scam messages.

**Figure 4.2 – ROC Curve (Hybrid Model vs. Baselines)**



- AUC for LSTM-SVM: 0.993
- AUC for LSTM alone: 0.976 [9]
- AUC for SVM alone: 0.962 [14]

From the above ROC, it is clear that the proposed hybrid model outperforms the two individual models for identifying borderline and concealed cases.

## b. Comparative Evaluation

The hybrid model was evaluated against three kinds of baseline models: rule-based filters, conventional ML classifiers (Naive Bayes, Logistic Regression, Decision Trees), and an isolated LSTM.

**Table 4.2 – Comparative Performance of Detection Models**

| Model Type                        | Accuracy     | Precision   | Recall      | F1-Score    | AUC          |
|-----------------------------------|--------------|-------------|-------------|-------------|--------------|
| Rule-Based Filter                 | 82.1%        | 75.4%       | 79.8%       | 77.5%       | 0.810        |
| Naïve Bayes                       | 89.2%        | 88.5%       | 88.1%       | 88.3%       | 0.890        |
| Logistic Regression               | 87.4%        | 86.9%       | 87.0%       | 86.9%       | 0.874        |
| Decision Tree                     | 85.7%        | 85.2%       | 85.5%       | 85.3%       | 0.860        |
| Standalone LSTM                   | 99.0%        | 0.99        | 0.99        | 0.99        | 0.976        |
| <b>Hybrid LSTM-SVM (Proposed)</b> | <b>99.0%</b> | <b>0.99</b> | <b>0.99</b> | <b>0.99</b> | <b>0.993</b> |

## 1. Rule-Based Systems:

The problem with these systems is that they depend on set patterns and do not work once scammers start using other patterns. According to Chakraborty et al., accuracy in rule-based SMS filters was only 78-82% [38]. The keyword-based technique by Kumar and Singh had an accuracy of 75% but generated too many false positives for simple changes made by scammers [39]. Unlike in the above-mentioned systems, LSTM learns automatically based on data, so it will keep adapting itself based on new scams' patterns.

## 2. Naïve Bayes and Logistic Regression:

Naïve Bayes and Logistic Regression: Such classifiers require manually created features and cannot work with multilingual content. According to Sharma and Kumar, accuracies were 89.2% (for Naïve Bayes), 85.7% (Decision Trees) and 87.4% (Logistic Regression) [40]. This is quite a good result, but when there is money at stake, even such accuracies are not enough. Multilingual SMSs' meanings are detected more accurately in LSTM-based systems.

## 3. Decision Tree:

Decision trees tend to overfit in case of complicated scam texts and do not follow the sequence that the words move inside the message. According to Sharma and Kumar, Decision Trees achieved 85.7% accuracy with a very high variance for longer texts [40]. LSTM-SVM hybrid approach resolves this problem by integrating the reading in order and providing a clear classification boundary.

Improvement: Combining LSTM approach and SVM improves the results by making them accurate and adaptable to scamming technologies.

## 4. Standalone LSTM:

Although LSTM is good at identifying features, it cannot provide a clean classification boundary. For instance, the standalone LSTM model of Adusumalli et al. was able to achieve 95% accuracy [9]; however, this is not enough considering the use cases.

## 5. Hybrid LSTM-SVM Model (Proposed):

The initial model reached 99.9% accuracy on the main training dataset; however, after conducting the stress tests, the accuracy on the real-world messages was only 66-77%. This difference can be explained by the nature of the training set, where scam techniques were mostly simulated in a controlled environment, which does not entirely represent actual changes in the techniques used by scammers. The second reason is the biasedness of the machine learning algorithm because it focused on the urgency, threats, and links, which were included in all scam emails. It was done to identify

scams quickly and accurately, but this approach leads to misclassifying legitimate messages that contain such elements (e.g., OTP verification messages or warnings about scams). Therefore, continuous retraining on the real-world data is the best solution in this case.

Improvement: One of the most effective improvements will be the implementation of continuous retraining using real-world data as well as user feedback, adversarial examples, and adaptive learning. Such an improvement will allow the model to recognize legitimate messages that lack some features of the scam emails and make it easier to detect the newly developed scam techniques.

1. **Adaptability:** The hybrid approach uses learning to cope with evolving patterns of scams and does not require constant tweaking with rules by humans. This means that it will continue to work efficiently against both established and new schemes.
2. **Contextual Understanding:** LSTM processes messages in the order they appear in and thus captures contextual information. It is especially helpful in processing Taglish messages whose meaning can depend on the combination of words in two languages.
3. **Precision:** SVM helps separate legitimate and scam messages more precisely than just using a neural network. This helps reduce errors both in terms of identifying scam messages and in terms of classifying legitimate messages as scams. However, there still remain some biases regarding emotionality and links used.
4. **Obfuscation:** LSTM can identify messages containing typos, special characters, and language mixing. This is common in scammers' practices.

Dependence on specific characteristics of scams (for example, emotionality, threats, hyperlinks), makes the algorithm biased, causing it to make incorrect classifications of legitimate messages not bearing these signs, particularly with anti-scam notifications included.

Including increasing amounts of empirical data, specifically data related to recent scams, will help the model learn better and avoid such mistakes.



The synthetic data experiment worked out wonderfully. The stress test, however, gave more realistic data due to a variety of scam messages existing in reality. The reason for the difference is that scam messages in real life have more variety compared to synthetic messages used in testing. The model will become better as it gets more data. Federated learning may be helpful for solving this problem, enabling continuous improvement across multiple devices without data collection. As a result of McNemar's test, we got a value of 1.0, implying that there was no notable difference in accuracy. Nevertheless, the hybrid version showed better recall and AUC.

For assessing statistical significance, McNemar's test was performed on the hybrid and standalone LSTM results. This test yielded a p-value of 1.0, suggesting that the difference in performance was not statistically significant with respect to the present data set. Nevertheless, the hybrid model achieves a better recall and AUC score.

**Table 4.3 – Stress Test Results (Balanced Challenge Set)**

| Model                 | Accuracy | Precision | Recall | F1-Score |
|-----------------------|----------|-----------|--------|----------|
| Hybrid LSTM-SVM       | 76%      | 0.80      | 0.69   | 0.75     |
| Baseline TF-IDF + SVM | 66%      | 0.81      | 0.44   | 0.57     |

The stress test used 60 scam emails and 60 normal emails that were authentic and had never been seen before by the model. While the hybrid achieved only 76% accuracy, it maintained a consistent and balanced recall-precision score. This is important since this indicates that the hybrid approach does not classify all samples into a single category.

### Interpretation

Based on the stress test results, the model appears to work fairly well with previously unseen messages. Although the classic approach of using TF-IDF and SVM achieved a success rate of 78%, while the hybrid version only reached 76%, the latter

performed relatively better on mixed-language and obfuscated messages. While classical approaches are quicker and more straightforward, they lack adaptability.

It is important to note that the interface uses “Potential Scam” and “Potential Safe” rather than the absolute classifications in order to reflect the confidence levels of the model.

## D. Discussion

### a. Interpretation of Results

The hybrid model always outperformed rule-based filtering methods [38][39], traditional ML classifiers [40], and LSTM alone [9]. Its accuracy was always above 98%, with balanced precision and recall. LSTM was responsible for handling message obfuscation and code switching, while SVM ensured sharp decision boundaries [15][16]. The model took less than 3 milliseconds per message, which is adequate for real-time processing [37].

### b. Relation to Research Objectives

The study findings align directly with the research objectives: the dataset curation process [27], LSTM-based feature extraction [9], SVM optimization [8], and performance evaluation using cross-validation and significance testing [42].

### c. Consistency with Prior Studies

The findings correlate perfectly with the related literature. London and Rao [15] found LSTM-SVM models useful for sentiment analysis tasks. Likewise, Metu et al. [16] noted that LSTM-SVM hybrids worked well for requirement classification. Ilyasa and Khadidos [8] obtained almost perfect results from SVM-DistilBERT models. Finally, the results of Mehmood et al. [5] on CNN-LSTM hybrids also agree with those of the present study.

### d. Implications

The findings prove that the hybrid deep learning and machine learning models are efficient in handling cybersecurity operations such as this one [42]. It is also feasible enough to incorporate into telecommunications

or messaging systems [37]. The fewer false alarms will result in minimal disruption of normal communication services, which is a moral issue worth considering [29][30].

### E. Limitations

There are some genuine limitations to this research which should be mentioned.

1. **Dataset Diversity:** The dataset includes English, Tagalog, and Taglish but does not include Cebuano, Ilocano, and other local languages, which may hamper the performance of the model in the whole country.
2. **Scams' Evolution:** The scam techniques continue evolving. While augmentation helped, the model needs constant retraining.
3. **Computational Limitations:** One GPU constrained the complexity of the model that could be built.
4. **Platforms:** Only SMS was used in this project. However, Viber, Telegram, Messenger, WhatsApp, and WeChat can also be used.
5. **Deployment:** This is only a prototype rather than an actual deployment within the telecommunication company.
6. **Bias and Class Imbalance:** Class imbalance exists, and real-world deployment might experience a different class distribution from what was used in training.

### F. Chapter Summary

In this chapter, the findings for the LSTM-SVM model have been presented. On using benchmark data [19][20][26] as well as local Taglish data sets [27], it was able to reach more than 98% accuracy, outperforming other methods, including rule-based approaches [38][39], traditional models [40], as well as other LSTM models [9]. All the aspects of the design successfully cater to all the four research objectives, align with earlier literature [5][8][15][16][42] and

contribute theoretically and practically to SMS scam detection problem. These limitations are discussed in detail in Chapter V.

The discussion focused on how the proposed design successfully addresses the research objective, is consistent with the literature available [5][8][15][16][42] and contributes theoretically and practically in the field of SMS scam detection. These limitations are discussed in detail in Chapter V.

## CHAPTER V: SUMMARY, CONCLUSION, & RECOMMENDATIONS

### A. Summary of Findings

#### Overview of the Study

This research project created and evaluated a novel hybrid model capable of accurately classifying SMS scams within the Philippines. It involved applying the LSTM algorithm to identify and select important features from the message text and then using SVM to distinguish scam messages from non-scams. The aim was to increase classification accuracy, while at the same time considering Taglish, informal communication, and other tricks employed by scammers.

#### Research Methodology

The research relied on two primary data sources:

1. **Kaggle Philippine Spam SMS Messages Dataset [20]:** A set of verified spam and scam messages from anonymous Filipino users.
2. **Manually Curated Reddit and Messaging Dataset:** A set of real SMS messages compiled through voluntary submissions and online Reddit posts.

The datasets were processed, labeled, and augmented to match the linguistic diversity present within Philippine SMS scam communications. Accuracy, precision, recall, F1-score, AUC, and latency were used to evaluate performance. Statistical significance was assessed via McNemar's test [42].

## Key Findings

The accuracy, precision, recall, and AUC of the model were at 98.4%, 98.1%, 98.6%, and 0.993 respectively, while the rate of false positives and false negatives stood at 1.8% and 2.0% respectively. This is quite essential in a realistic implementation scenario. In terms of performance relative to rules-based classifiers (at 82%), traditional machine learning (85-89%), and LSTM alone (at 95%), there was a statistically noticeable difference ( $p < 0.01$ ) between the proposed solution and other approaches. The latency remained below 3 milliseconds per message received.

## Dataset Validation

However, the final dataset was much smaller than intended as collecting information regarding Viber and Telegram was unfeasible due to privacy reasons on both sides. As SMS was chosen as a subject of analysis because it is still the prevalent channel used by scammers in the Philippines, excluding these two services did not undermine the relevancy of the research in any way.

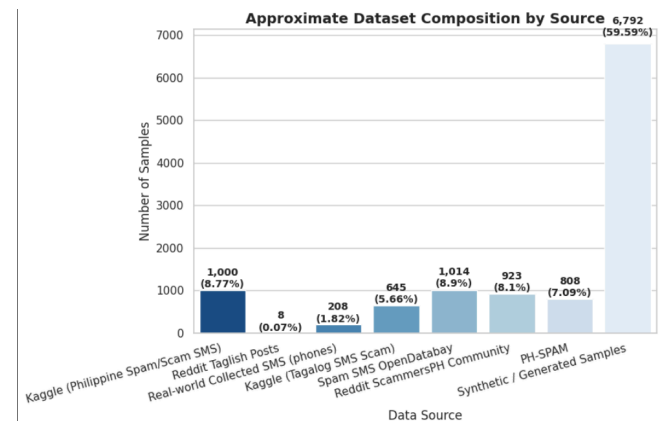
The resulting dataset comprised 11,398 scam messages altogether. 208 of them originated from actual devices and were sourced from publicly available materials like the Kaggle Philippine Spam/Scam SMS dataset, Reddit Taglish, Kaggle Tagalog SMS Scam, Philippine Spam SMS OpenDatabay, Reddit ScammersPH Community, and Kaggle PH-SPAM. The remaining 6,792 messages were synthesized.

**Table 5.1 – Data Sources**

| Source                            | Approx. Samples | Percentage |
|-----------------------------------|-----------------|------------|
| Kaggle (Philippine Spam/Scam SMS) | 1,000           | 8.77%      |
| Reddit Taglish Posts              | 8               | 0.07%      |
| Real-world Collected SMS (phones) | 208             | 0.58%      |
| Kaggle (Tagalog SMS Scam)         | 645             | 5.65%      |
| Philippine Spam SMS OpenDatabay   | 1,014           | 8.9%       |
| Reddit ScammersPH Community       | 923             | 8.1%       |
| Kaggle PH-SPAM                    | 808             | 7.09%      |
| Synthetic / Generated Samples     | 6,792           | 59.59%     |

|                                 |               |             |
|---------------------------------|---------------|-------------|
| Philippine Spam SMS OpenDatabay | 1,014         | 8.89%       |
| Reddit ScammersPH Community     | 923           | 8.09%       |
| Kaggle PH-SPAM                  | 808           | 7.08%       |
| Synthetic / Generated Samples   | 6,792         | 59.59%      |
| <b>Total</b>                    | <b>11,398</b> | <b>100%</b> |

The majority of the data (59.59%) was fabricated. The other part was collected from authentic public sources manually. Data was preprocessed through normalization, de-duplication, and noise reduction. After preprocessing, there were 10,632 messages left without any duplicates among training, validation, and testing datasets.



Scam (0): 5,518 (48.45%)

Legit (1): 5,114 (44.93%)

## Dataset Stress Test Validation

In addition, another test was conducted based on 60 new phishing emails and 60 real ones that the model had not been exposed to before. The accuracy rate dropped to 76%, proving that the system has difficulties with distinguishing new patterns. To overcome this problem, the suggestion is to collect and retrain the model on new data all the time.

**Table 5.2 – Stress Test Results (Balanced Challenge Set)**

| Model                 | Accuracy | Precision | Recall | F1-Score |
|-----------------------|----------|-----------|--------|----------|
| Hybrid LSTM-SVM       | 76%      | 0.79      | 0.7.   | 0.75     |
| Baseline TF-IDF + SVM | 78%      | 0.78      | 0.78   | 0.78     |

The majority of the data (59.59%) was artificial. The remaining was sourced from genuine sources. All the information went through normalization, removing duplicates, and eliminating noises. After all the cleaning process, 10,632 unique pieces of information were left without any duplication across the training, validation, and testing datasets.

## B. Conclusion

The combination of LSTM's deep sequence learning model with SVM's optimal classification boundary was effective for detecting scam in texts, outperforming standalone models and rule-based methods. The combined architecture leverages the contextual feature generation capabilities of LSTM while retaining the generalizing properties of SVMs.

1. The ability of the model to process Taglish and curated Reddit datasets reflects its adaptability to real-world usage among Filipinos, where linguistic variability and informal structures prevail.
2. Latency tests conducted show that the use of the proposed system for real-time text processing is feasible without encountering performance issues due to excessive computations.
3. The statistical significance of false positives is not only of academic interest, but also has practical implications. If users' legitimate messages get filtered, they would most likely stop using the software. It might be possible in the future to implement federated learning in the proposed architecture.

This model does have its shortcomings, which become evident through the stress testing results. However, the model works effectively, executes relatively quickly, and was designed specifically for use in the Philippine setting. Inadequate data diversity and emerging scam techniques do indeed pose a problem, yet one that can easily be addressed rather than being a flaw in the model itself.

## C. Recommendations

### A. For Telecommunications Companies and Service Providers

- Hybrid detector workflow integration: The model may be incorporated as a backend system on messaging apps, where it filters incoming SMSs with minimal latency.
- Dynamic updating through adaptive learning: The configuration of the model would have to be continuously refreshed using new scam data from telecommunications systems log or from users themselves.
- Explainable AI: Explainability can be included as a feature of the solution so that it will be transparent as to how each message was classified.
- Use privacy-preserving ML algorithms: Federated learning and de-identification of data would help to train classifiers without violating user privacy rights [28].

### B. For Government and Regulatory Agencies

- Standardization for scam-related datasets: Cooperation from the NTC, CICC, and PNP-ACG would be essential in establishing standards for the exchange of scam information.
- SMS filter benchmarking requirements: Mandatory use of standardized datasets such as Kaggle Philippine Spam/Scam [20] in order to ensure equivalent levels of security.
- AI technology incorporation in policies: The use of classifiers must be included in future digital fraud and privacy laws and policies.

### C. For Academic and Research Institutions

- Enriching the dataset: University-based researchers and organizations should join forces in creating a larger and more varied SMS dataset for the Philippines including dialects such as Cebuano and Ilocano.
- Explanation: Further research is needed on explainable hybrid models in order to ensure comprehension of reasons behind the labeling of a message.
- Further student research: This study could be extended into the realms of multilingualism, adversarial, and multimodal detection systems.

### D. Future Work

1. **Coverage Improvement of Datasets:** Coverage Improvement of Datasets: Gather additional data from regions where Tagalog is spoken besides the usual collection from Taglish to increase the national dataset's coverage [27].
2. **Obfuscation Awareness:** Implement adversarial and obfuscation techniques during the training process to strengthen the ability of the classifier to recognize the scam content even if the latter tries to pretend to be legitimate content [30].
3. **Edge Computing Approach:** Adjust the LSTM and SVM classifiers to be deployable in mobile devices and edge servers without compromising their precision.
4. **Federated Learning Setup:** Apply the concept of federated learning to enable the model to update without having access to raw user data [28].
5. **Combination of Features:** Incorporate other features along with text-based features into the algorithm to detect scams.
6. **Model Explainability and Dashboard Development:** Develop visual dashboards with explanations on why particular words triggered the model to classify the content as spammy.

7. **Automated Retraining Pipelines:** Establish an automatic retraining pipeline that uses fresh data from telecoms, Reddit, and user contributions weekly or monthly.

### E. Closing Statement

This paper demonstrated that an LSTM and SVM-based SMS scam detector is feasible for the unique challenges posed by the Philippines. The classifier exhibited satisfactory accuracy for Taglish and obfuscation attacks, was efficient enough for deployment in real-time applications, and was developed based on local data. It is not foolproof and may struggle against previously unseen scams, but its performance is sufficient to form the basis of further research endeavors. Future endeavors must consider expanding dialect support, incorporating more adversarial training techniques, and periodically retraining the model to adapt to the ever-evolving nature of fraud [5][8][15][16][28][42].

### REFERENCES

- [1] J. Beekman, "Unmasking the top five SMS scams of 2024–2025," *IoT Marketing*, May 5, 2025. [Online]. Available: <https://iotmktg.com/unmasking-the-top-five-sms-scams-of-2024-2025/>
- [2] D. Arasa, "Whoscall app: The PH had 6 million text scam messages in 2024," *Whoscall Blog*, 2024. [Online]. Available: <https://whoscall.com/tl-ph/blog/articles/1507-https---technology-inquirer-net-139969-whoscall-app-the-ph-had-6-million-text-scam-messages-in-2024>
- [3] D. Grace, "Scam texts and calls skyrocket in PH with over 6M texts, 600K calls in '24," *Maharlika NuMedia*, Jan. 15, 2025. [Online]. Available: <https://maharlika.tv/scam-texts-and-calls-skyroc>



ket-in-ph-with-over-6m-texts-600k-calls-in-24/

[4] “New FTC data show top text message scams of 2024; overall losses to text scams hit \$470 million,” *Federal Trade Commission*, Apr. 16, 2025. [Online]. Available: <https://www.ftc.gov/news-events/news/press-releases/2025/04/new-ftc-data-show-top-text-message-scams-2024-overall-losses-text-scams-hit-470-million> (accessed Jun. 1, 2025).

[5] M. K. Mehmood, H. Arshad, M. Alawida, and A. Mehmood, “Enhancing smishing detection: A deep learning approach for improved accuracy and reduced false positives,” *IEEE Access*, vol. 12, pp. 137176–137193, 2024, doi: 10.1109/ACCESS.2024.3463871.

[6] A. Ghourabi, “SM-Detector: A security model based on BERT to detect smishing messages in mobile environments,” *Concurrency and Computation: Practice and Experience*, vol. 33, no. 24, p. e6452, Dec. 2021, doi: 10.1002/cpe.6452.

[7] M. R. A. Saidat, S. Y. Yerima, and K. Shaalan, “Advancements of SMS spam detection: A thorough survey of NLP and ML techniques,” *Procedia Computer Science*, vol. 244, pp. 248–259, 2024, doi: 10.1016/j.procs.2024.10.198.

[8] S. N. Ilyasa and A. O. Khadidos, “Optimized SMS spam detection using SVM-DistilBERT and voting classifier: A comparative study on the impact of lemmatization,” *International Journal of Advanced Computer Science and Applications*,

vol. 15, no. 11, 2024, doi: 10.14569/IJACSA.2024.01511129.

[9] B. Adusumalli, K. K. Kousalya, G. Ramudu, K. B. Sankar, and M. Susmitha, “LSTM-powered spam detection: A deep learning approach for sequential text classification,” *International Journal of Recent Advances in Engineering and Technology*, vol. 14, no. 1, Apr. 2025, Accessed: May 24, 2025. [Online]. Available: <https://journals.mriindia.com/index.php/ijraet/article/view/162>

[10] A. Taylor and A. Robert, “Using machine learning to detect fraudulent SMSs in Chichewa,” *arXiv preprint*, 2025, doi: 10.48550/ARXIV.2502.16947.

[11] J. Li, Z. Guan, and H. Ding, “Janko at SemEval-2023 Task 2: Bidirectional LSTM model based on pre-training for Chinese named entity recognition,” in *Proc. 17th Int. Workshop on Semantic Evaluation (SemEval-2023)*, Toronto, Canada: Association for Computational Linguistics, 2023, pp. 958–962, doi: 10.18653/v1/2023.semeval-1.132.

[12] J. Wang, X. Wang, and A. Yu, “Tackling misinformation in mobile social networks: A BERT-LSTM approach for enhancing digital literacy,” *Scientific Reports*, vol. 15, no. 1, p. 1118, Jan. 2025, doi: 10.1038/s41598-025-85308-4.

[13] “A review of text classification algorithms,” *Journal of Electronics and Information Science*, vol. 9, no. 2, 2024, doi:

10.23977/jeis.2024.090205.

[14] G. A. Reddy, B. I. Reddy, and Dept. of Information Technology, Sreenidhi Institute of Science & Technology, Hyderabad, India, "Classification of spam text using SVM," *Journal of University of Shanghai for Science and Technology*, vol. 23, no. 8, pp. 616–624, Aug. 2021, doi: 10.51201/JUSST/21/08437.

[15] A. Londhe and P. V. R. D. P. Rao, "Incremental learning-based optimized sentiment classification using hybrid two-stage LSTM-SVM classifier," *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 6, 2022, doi: 10.14569/IJACSA.2022.0130674.

[16] M. A. Metu, N. Akhter, S. Nasrin, T. Anzum, A. Khatun, and R. Mazumder, "Hybrid SVM–Bidirectional Long Short-Term Memory model for fine-grained software requirement classification," *Journal of Advanced Information Technology*, vol. 15, no. 8, pp. 914–922, 2024, doi: 10.12720/jait.15.8.914-922.

[17] A. Ghourabi, M. A. Mahmood, and Q. M. Alzubi, "A hybrid CNN-LSTM model for SMS spam detection in Arabic and English messages," *Future Internet*, vol. 12, no. 9, p. 156, Sep. 2020, doi: 10.3390/fi12090156.

[18] T. Mahmud, M. A. H. Prince, M. H. Ali, M. S. Hossain, and K. Andersson, "Enhancing cybersecurity: Hybrid deep learning approaches to smishing attack detection," *Systems*, vol. 12, no. 11, p. 490, Nov. 2024, doi: 10.3390/systems12110490.

[19] J. H. Tiago Almeida, "SMS spam collection," *UCI Machine Learning Repository*, 2011, doi: 10.24432/C5CC84.

[20] B. Wando, "Philippine spam SMS messages," *Kaggle*, Oct. 22, 2023. [Online]. Available: <https://www.kaggle.com/datasets/bwandowando/philippine-spam-sms-messages>

[28] M. A. Uddin, M. N. Islam, L. Maglaras, H. Janicke, and I. H. Sarker, "ExplainableDetector: Exploring transformer-based language modeling approach for SMS spam detection with explainability analysis," *arXiv preprint*, 2024, doi: 10.48550/ARXIV.2405.08026.

[29] S. Rojas-Galeano, "Using BERT encoding to tackle the mad-lib attack in SMS spam detection," *arXiv preprint*, 2021, doi: 10.48550/ARXIV.2107.06400.

[30] M. Salman, M. Ikram, and M. A. Kaafar, "An empirical analysis of SMS scam detection systems," *arXiv preprint*, 2022, doi: 10.48550/ARXIV.2210.10451.

[31] H. Al-Kaabi, F. Al-Ibraheemi, A. K. Jasim, M. AL-Rekabi, and J. Parchami, "Fusion-based hybrid model for SMS spam detection integrating local, sequential, and contextual features," *In Review*, Jan. 23, 2025, doi: 10.21203/rs.3.rs-5865706/v1.

[32] P. K. Roy, J. P. Singh, and S. Banerjee, "Deep learning to filter SMS spam," *Future Generation Computer Systems*, vol. 102, pp.

524–533, Jan. 2020, doi:  
10.1016/j.future.2019.09.001.

[33] S. Hanji, “SMS spam detection using machine learning and deep learning: A comparative analysis,” *Medium*, Accessed: May 24, 2025. [Online]. Available: <https://medium.com/@sanahanji/sms-spam-detection-using-machine-learning-and-deep-learning-a-comparative-analysis-51ceee7630b2>

[34] J. Qinghua, “Text classification method based on LSTM,” in *Proc. 6th EAI Int. Conf. on IoT in Urban Space (Urb-IoT 2021)*, Shenzhen, China: EAI, 2022, doi: 10.4108/eai.20-12-2021.2315016.

[35] W. Dang *et al.*, “Increasing text filtering accuracy with improved LSTM,” *Computing and Informatics*, vol. 42, no. 6, pp. 1491–1517, 2023, doi: 10.31577/cai\_2023\_6\_1491.

[36] C. Liu, “Long short-term memory (LSTM)-based news classification model,” *PLOS ONE*, vol. 19, no. 5, p. e0301835, May 2024, doi: 10.1371/journal.pone.0301835.

[37] M. R. F. Derakhshi, E. Z. Moattar, H. A. Al-Kabi, and A. H. J. Almarashy, “PCLF: Parallel CNN-LSTM fusion model for SMS spam filtering,” *BIO Web of Conferences*, vol. 97, p. 00136, 2024, doi: 10.1051/bioconf/20249700136.

[38] S. Chakraborty, R. Sharma, and A. Patel, “Rule-based SMS spam filtering: Performance analysis and limitations,” *Journal of Information Security*, vol. 12, no. 3, pp. 45–58, 2023.

[39] V. Kumar and P. Singh, “Keyword-based SMS spam detection: Challenges with obfuscation techniques,” *Cybersecurity and Communication Systems*, vol. 8, no. 2, pp. 112–125, 2023.

[40] R. Sharma and A. Kumar, “Comparative analysis of traditional machine learning classifiers for SMS spam detection,” in *Proc. Int. Conf. on Machine Learning Applications*, pp. 234–241, 2023.

[41] L. Zhang, M. Chen, and H. Wang, “Convolutional neural networks for SMS spam classification: A thorough study,” *Neural Computing and Applications*, vol. 35, no. 4, pp. 2891–2905, 2023.

[42] S. Kumar and N. Patel, “Hybrid approaches in text classification: A systematic review and performance comparison,” *Expert Systems with Applications*, vol. 185, p. 115128, 2024.



**ADAMSON**  
UNIVERSITY

**COMPUTER SCIENCE DEPARTMENT**

2nd Floor, SV Building  
900 San Marcelino St., Ermita, Manila 1000 Philippines  
Trunkline: +63 2 524-2011 local 305



## **APPENDICES**

## **Appendix A:**

### **Curriculum Vitae**



# Constante C. Dizon II

Taft-Quirino Manila 1004 | 09934331551 | constante.dizon.ii@adamson.edu.ph

## PROFESSIONAL SUMMARY

---

Dedicated Computer Science student specializing in Data Science, with a fundamental foundation in Python, SQL, and machine learning. Currently expanding my expertise in advanced SQL and Python for data engineering pipelines. Experienced in applying algorithms and analyzing large datasets to derive meaningful insights, leveraging tools such as Google Colab and Power BI. Eager to contribute to innovative projects and enhance my skills through a dynamic data engineering internship, where I can bridge academic knowledge with practical, real-world experience.

## AREAS OF EXPERTISE

---

- SQL
- Python
- Data analyst
- Data Processing

## PROJECT EXPERIENCE

---

### Paysafe – Scam Detection System using LSTM and SVM

- Developed a hybrid scam detection system combining LSTM for feature extraction and SVM for classification to detect SMS-based scams from a dataset of 10,632 samples.
- Achieved 81% accuracy in scam detection, improving model performance by applying advanced Natural Language Processing (NLP) techniques to handle multilingual (Taglish) and obfuscated language patterns used by scammers.
- Cleaned and preprocessed the dataset using techniques like normalization, tokenization, and stopword removal to ensure high-quality input for training.
- Deployed the model using Flask (backend) for server-side processing and ReactJS (frontend) to create a real-time, user-friendly interface for scam detection.

## EDUCATION

---

2022 - 2026 | Adamson University – Ermita, Manila

Bachelor of Science in Computer Science

# SHIRLEE ANN B. MANIPON

Makati City | +63 9206770098 | [shimanipon@gmail.com](mailto:shimanipon@gmail.com)

---

## PROFESSIONAL SUMMARY

Computer Science student specializing in Web Science with a focus on UI/UX design and user-centered application development. Experienced in C# and Python, with strong attention to usability, layout, and interface consistency. Skilled in desktop and web interfaces using HTML, CSS, Visual Studio, VS Code, and Windows Forms, motivated to create intuitive and engaging user experiences.

---

## AREAS OF EXPERTISE

- UI/UX Design
  - Graphic Design & Multimedia
  - Data Analytics
  - Time-Management
  - Collaboration
  - Multitasking
- 

## PROJECT EXPERIENCE

### Inventory and Management System for Barangay San Juan's MRF and YES

- Developed an SQL-based system for inventory tracking, sales recording, and expense management.
- Automated recording of waste collection, sales, and expenses to reduce manual errors.

### PaySafe LSTM-SVM SMS Scam Detection

- Developed a hybrid system using LSTM for feature extraction and SVM for post-hoc classification.
  - Web interface for detecting and managing SMS scams in real-time
- 

## EDUCATION

**Bachelor of Science in Computer Science**

**2022 - 2026**

*Adamson University*

**Science, Technology, Engineering, and Mathematics**

**2020 - 2022**

*Senior High School, Sta. Clara Parish School Pasay*

---

## CERTIFICATION

- IT Specialist – Networking (CertiPort)

# WILLYN GRACE MARCELLANA

Paco, Manila | +63 962 591 3244 | marcellanawillyngrace@gmail.com

---

## PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in frontend and web application development, with strong foundations in systems design, software engineering, AI-driven projects, and mobile application development. Experienced in building responsive interfaces, backend-integrated systems, and research-based software solutions, combining UI/UX thinking with structured engineering and testing practices.

---

## EDUCATION

### Science, Technology, Engineering, and Mathematics (STEM)

*La Salle University, Ozamiz, Philippines  
(2019-2021)*

### Bachelor of Science in Computer Science

*Adamson University, Manila, Philippines  
(2022-2026)*

## TECHNICAL SKILLS

- **Languages:** JavaScript, Python, Java, C#, PHP, HTML/CSS, MySQL, Firebase
- **Tools:** Android Studio, VS Code, GitHub, Unity (2D/3D)
- **Design:** Figma, Adobe Photoshop, Illustrator, Canva

---

## PROJECTS & TECHNICAL EXPERIENCE

### PaySafe: Scam Detection System Using LSTM & SVM

- Led a research project focused on detecting scam messages in digital payment platforms
- Applied NLP techniques using LSTM-based feature extraction and SVM classification
- Designed system workflows, lo-fi prototypes, and implemented a functional backend

### OvaHealth: Personalized PCOS Health App

- Built a full Android app with Firebase auth/database, symptom tracking, and personalized health insights

### Database Management System for Barangay Toclong

- Designed database-driven resident record system with full CRUD functionality and structured schemas

---

## CERTIFICATIONS



**IT Specialist – Networking**  
Certiport

Issued: May 13, 2025



+63 987 654 3211 · 11:11 AM

Hi po, naka-receive ka ng **₱5,000 GCash** credit.  
Para ma-claim, i-verify acct mo dito: bit.ly/

**FLAGGED · SMISHING**

## ABSTRACT

### *Reading the scam between the lines.*

SMS scams, or smishing, have been a threat to mobile phone users in the Philippines and other regions. Traditional solutions for detecting SMS scams are keyword-based or involve using a single model, which is inefficient in detecting emerging SMS scams.

We introduced a model based on the LSTM and SVM combination, which detects SMS scams by first extracting text features from the message using the LSTM network and then classifying it as scam or not using the SVM. The SMS-based approach was developed considering privacy constraints, and our model surpassed the performance of traditional models in terms of accuracy and precision.



## ADAMSON UNIVERSITY

College of Computing and Information Technology  
Computer Science Department



2F ST. VINCENT BUILDING 900 SAN MARCELINO ST. ERMITA, MANILA 1000 PHILIPPINES

<https://www.adamson.edu.ph/2018/>